Audio-Agent: Bridging the Semantic Gap in Neural Audio Effects via Multi-Modal Retrieval-Augmented Generation

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***Abstract*—Modern Digital Audio Workstations (DAWs) expose rich effect chains, but a semantic gap remains between perceptual user intent (e.g., "warm", "aggressive", "spacious") and low-level DSP control parameters. Recent generative models can synthesize waveforms directly, yet waveform outputs are difficult to edit in production workflows. We present Audio-Agent, a neuro-symbolic framework that generates *editable DSP parameter configurations* instead of finalized waveforms.**

Audio-Agent combines three components: (1) Texture Resonance Retrieval (TRR), which encodes audio texture with second-order feature statistics (Gram matrices); (2) Dual-Modal Adaptive Fusion, which reweights text and audio evidence using retrieval uncertainty; and (3) Constrained Parameter Projection, which deterministically repairs LLM drafts to satisfy plugin feasibility constraints.

In a pilot study on a server-side guitar-effects corpus of 1082 paired records, we report objective parameter-space evaluation on a fixed 30-query held-out split (1052-item retrieval knowledge base). Under this protocol, TRR achieves lower parameter distance than classical texture baselines (13.57 vs. 24.95 for MFCC temporal statistics and 32.02 for modulation spectrogram). Compared to pure LLM generation without retrieval (Param. Dist.=13.36), our retrieval-grounded approach reduces parameter error by 97.7% (Param. Dist.=0.30), demonstrating the necessity of retrieval augmentation. Uncertainty-aware fusion further reduces parameter distance from 13.15 (best single modality) to 0.93. We complement objective evaluation with a MUSHRA-style listening study (26 participants, 910 ratings). All quantitative claims are reported as split-specific evidence under the stated protocol without broader universal generalization claims.

I. Introduction

The field of neural audio synthesis currently faces a fundamental tension between *fidelity* and *control*. Recent high-fidelity generative models, primarily based on latent diffusion [1, 2], can achieve strong perceptual quality but often operate as "black boxes." They generate finalized waveforms where internal attributes—such as the attack time of a compressor or the decay of a reverb—are entangled and inaccessible for post-hoc editing. Conversely, Differentiable Digital Signal Processing (DDSP) [3] offers interpretable, structured control but often suffers from the ill-posed nature of the inverse parameter estimation problem.

This tension motivates the development of **Neuro-Symbolic Audio Agents**: systems that leverage the reasoning capabilities of Large Language Models (LLMs) to control interpretable DSP engines. By predicting parameters rather than samples, such systems ensure that outputs are not just high-fidelity, but also fully editable within professional production workflows.

However, realizing this paradigm requires overcoming two critical challenges. First, LLMs trained on discrete tokens struggle to regress continuous values within strict physical constraints, leading to a *hallucination problem* where generated parameters violate DSP valid ranges (e.g., negative filter frequencies) or inter-parameter dependencies. Second, grounding the LLM via retrieval requires a similarity metric that captures *timbral texture*. Standard audio representations (e.g., CLAP-style embeddings or mean-pooled Wav2Vec2 features) summarize an audio segment into a single vector; while effective for coarse semantics, such pooling can discard temporal co-activation patterns that are critical for texture-dominant effects (e.g., tremolo modulation or signal-dependent distortion).

To bridge these gaps, we present **Audio-Agent**, a framework that frames parameter generation as a *retrieval-grounded reasoning task*. Our system retrieves validated presets based on user intent and adapts them via an LLM to satisfy user queries. To address the texture representation gap, we draw inspiration from the neuroscience of sound texture perception [4] and visual style transfer [5], introducing **Texture Resonance Retrieval (TRR)**. TRR models audio style using Gram matrices of deep feature activations, explicitly capturing the *second-order correlations* that define temporal texture.

Our key contributions are: (i) **Texture-Aware Grounding**: we observe that second-order statistics (Gram matrices) can improve retrieval for texture-dominant effects, yielding better parameter alignment on our benchmark; (ii) **Uncertainty-Aware Modal Fusion**: we propose an entropy-based fusion mechanism that adjusts the relative weight of text and audio retrieval under degraded inputs; and (iii) **Constrained Neuro-Symbolic Execution**: we introduce a deterministic constraint projection layer that enforces plugin-defined validity constraints on LLM-generated parameters.

II. Related Work

Our work lies at the intersection of intelligent music production, neural audio synthesis, and LLM-based agent systems. We review each area and position our contributions accordingly.

Intelligent Music Production.

The challenge of translating perceptual descriptions into technical parameters was formally characterized by Stables et al. [6]. Their SAFE project demonstrated the feasibility of semantic-to-parameter mapping in a DAW context, but did not resolve the generalization problem across instruments, effect chains, and stylistic sub-genres. Schedl et al. [7] provides a comprehensive survey of music information retrieval techniques, highlighting semantic auto-tagging as a key challenge.

Early work in interactive control focused on learning intuitive mappings for synthesizers. Kulić et al. [8] proposed active learning with Gaussian processes to discover perceptual control knobs, while more recently, Bryan et al. [9] introduced SynthScribe, a multimodal tool for synthesizer sound retrieval. While these works advance interface design, they primarily target synthesis parameters (e.g., oscillator shape) rather than the complex signal chains of audio effects.

Audio Analysis Approaches. Subsequent work pursued automation via audio content analysis and differentiable mixing systems [10–12]. While effective when a target mix or reference audio already exists, this direction implicitly assumes the target audio is given. It therefore struggles to support creative timbre design from a purely conceptual request ("make it more vintage breakup") and often suffers from an **opacity problem**: users see outputs but lack interpretable, editable rationale.

Recent advances in differentiable DSP have enabled neural approaches to audio effect modeling. Peladeau et al. [13] proposes blind estimation of audio effect parameters using auto-encoder architectures, addressing the inverse problem of parameter recovery. Liu et al. [14] introduces DDSP-SFX, which leverages differentiable digital signal processing for acoustically-guided sound effects generation, combining neural networks with traditional DSP modules for controllable synthesis. De Man et al. [15] explores generative latent spaces for neural audio synthesis, bridging DSP-informed and data-driven approaches. In parallel, style transfer methods have been applied to audio effects, with Benetos et al. [16] introducing ST-ITO for controlling audio effects via inference-time optimization using pretrained audio representations.

The LLM Frontier. Recent work explores LLMs as natural language interfaces for effect parameters [17–19]. While demonstrating surprising capability, these approaches remain vulnerable to hallucination: reported violation rates can reach **up to 35%** in zero-shot settings, producing invalid or incoherent parameters. Dialogue-based disambiguation can reduce ambiguity but shifts the burden to the user and does not scale to complex multi-effect chains.

Our Positioning. Unlike prior work that treats parameter prediction as a *regression problem* or a purely generative problem, we frame it as a *retrieval-grounded reasoning problem*: retrieval supplies concrete, previously validated exemplars, and generation becomes an adaptation step rather than invention.

Neural Audio Synthesis.

Neural audio synthesis has advanced rapidly, enabling high-fidelity text-to-audio generation [1, 2, 20, 21]. Diffusion models have emerged as the dominant paradigm, with Huang et al. [22] introducing prompt-enhanced text-to-audio generation, Wang et al. [23] achieving efficient high-quality synthesis via latent consistency models, and Liang et al. [24] exploring instruction-guided latent diffusion for better controllability. However, these models primarily output *finalized waveforms*, which are difficult to edit within professional workflows.

TABLE I: Comparison of waveform-level and parameter-level approaches in neural audio workflows. Each paradigm offers complementary strengths for different production contexts.

Level	Primary Strength	Primary Limitation
Waveform-level	High-fidelity synthesis	Hard to edit in DAWs
Parameter-level	Fully editable control	Requires expert priors

Accordingly, we view parameter-level generation as a **complement to, not a replacement for**, waveform-level synthesis: it targets controllable editing and iterative refinement inside DAWs.

This motivates what we call **Parameter–Waveform Duality**: waveform-level generation offers maximal expressivity but limited editability, whereas parameter-level control preserves transparency and iterative refinement at the cost of requiring domain knowledge.

In parallel, differentiable DSP (DDSP) [3] demonstrates that learned models can control interpretable synthesis parameters, and self-supervised audio representation learning (e.g., Wav2Vec 2.0 [25]) provides strong acoustic embeddings. Audio-Agent leverages these representations for similarity grounding, but targets editable parameter outputs compatible with DAW-centric workflows.

LLM Agents and Retrieval-Augmented Generation.

Retrieval-augmented generation (RAG) [26] improves reliability by grounding generation in external memory. Multimodal variants extend this idea beyond text [27, 28], with recent developments including MMKB-RAG [29] for multimodal knowledge-based systems, and frameworks for visually-rich document understanding [30]. Instruction manual understanding approaches [31] demonstrate the value of cross-modal grounding in practical applications. Agentic tool-use frameworks (e.g., [32–34]) enable language models to act through structured interfaces.

Although our focus is parameter generation, we acknowledge the rapid progress in neural audio synthesis. Recent models like AudioLDM 2 [20], AudioCraft [35], and industrial systems like Suno illustrate the power of end-to-end generation. However, these systems generally do not expose the *internal signal chain* for fine-grained editing, which remains the domain of parameter-based approaches.

This issue persists even when using strong general-purpose audio representations (e.g., CLAP- or PaSST-style embeddings): perceptual similarity does not necessarily imply that the underlying effect parameters are transferable. Recent audio-language models such as CLAP [36], ReCLAP [37], and M2D-CLAP [38] have improved zero-shot audio classification through better text-audio alignment, while few-shot prompt learning techniques [39] enable better adaptation to downstream tasks. However, these models primarily target classification and captioning rather than parameter estimation.

Audio Texture and Semantic Audio Retrieval.

The concept of audio texture has been studied extensively in the context of style transfer and synthesis. Gatys et al. [5] introduced Gram matrix-based style representations for images, which were later adapted to audio by Stylianou et al.

[40] and others. Recent surveys on audio texture analysis [41] and audio representations for sound synthesis [42] highlight the importance of capturing second-order statistical features for timbre characterization.

In semantic audio retrieval, the WikiMuTe dataset [43] provides a rich resource for connecting natural language descriptions with music audio, while retrieval-guided approaches [44] demonstrate the value of combining retrieval with generation for music captioning. Retrieval-augmented text-to-music generation [45] further validates the effectiveness of RAG approaches in audio tasks.

Our Texture Resonance Retrieval (TRR) builds upon these insights by using Gram matrices to capture audio texture through second-order statistics, enabling more style-consistent retrieval for parameter inference.

Table II summarizes our positioning. Our distinctive contribution is combining (1) *dual-modal* grounding for parameter generation, (2) *texture-aware* audio retrieval via TRR, and (3) *uncertainty-aware* fusion for robustness under degraded inputs.

III. Problem Formulation

We formalize neuro-symbolic audio control as a constrained parameter synthesis problem: given a user query and an input signal, produce a parameter vector that is (i) *semantically consistent* with the user's intent and (ii) *executable* by a real-time DSP engine. Unlike waveform generation, parameter synthesis must respect strict plugin-defined validity constraints, and the inverse mapping from perceptual language (e.g., “warm”, “punchy”) to a high-dimensional control vector is inherently ill-posed. Our goal is therefore not to claim a globally optimal solution in Θ, but to produce a feasible point estimate that is well-defined, interpretable, and grounded by concrete executable prototypes.

A. Signal Processing Dynamics

Consider a (potentially non-differentiable) DSP rendering operator $\Phi : \mathcal{X} \times \Theta \to \mathcal{X}$, where $\mathcal{X}$ denotes the space of discrete-time audio signals and $\Theta \subset \mathbb{R}^d$ denotes the d-dimensional control parameter space. Given an input signal $\mathbf{x}_{\text{in}} \in \mathcal{X}$, the operator produces an output $\mathbf{x}_{\text{out}} = \Phi(\mathbf{x}_{\text{in}}; \theta)$. In practical DAW and plugin systems, Φ can include nonlinearities, conditional branches (e.g., bypass logic), and internal state (e.g., smoothing), making end-to-end differentiation unreliable or inapplicable for direct optimization over θ.

The parameter space Θ is not a free vector space but a bounded hyper-rectangle subject to structural constraints:

$$\Theta = \left\{ \theta \in \mathbb{R}^d \mid \mathbf{l} \preceq \theta \preceq \mathbf{u}, \quad \mathcal{C}(\theta) = 1 \right\} \tag{1}$$

where $\mathbf{l}, \mathbf{u}$ define physical operating ranges (e.g., frequency bounds $[20, 20\text{k}]$), and $\mathcal{C} : \mathbb{R}^d \to \{0, 1\}$ represents a binary validity function encoding inter-parameter dependencies (e.g., ensuring filter resonance $Q > 0$). In practice, DSP parameter spaces can be non-convex and partially discrete (e.g., on/off switches or mode selectors), so we use Eq. 1 as a pragmatic executable abstraction rather than a claim of convexity or smoothness. This motivates our reliance on retrieval-grounded synthesis and deterministic validity enforcement instead of direct gradient-based optimization over Θ.

B. Goal and Constraints

The user's intent is encapsulated in a multi-modal query $q = (t, \mathbf{a}_{\text{ref}})$, comprising a natural language description $t \in \mathcal{T}$ and an optional audio reference $\mathbf{a}_{\text{ref}} \in \mathcal{X} \cup \{\emptyset\}$. Our objective is to produce a *valid* parameter configuration that is consistent with the query:

$$\tilde{\theta} \in \Theta \quad \text{given } q, \tag{2}$$

where validity is defined by the plugin-specific constraint function $\mathcal{C}(\theta) = 1$ and physical bounds $\mathbf{l} \preceq \theta \preceq \mathbf{u}$ (Eq. 1). We use θ^* to denote an ideal (possibly non-unique) solution that best matches the user's perceptual intent; in practice we only have access to imperfect retrieval and LLM reasoning, and therefore aim for a feasible point estimate $\tilde{\theta}$.

a) Operational objective (without probabilistic semantics).: The query-consistency of $\tilde{\theta}$ is not directly measurable as a differentiable loss, because the target is perceptual and depends on a complex signal chain. Accordingly, we adopt a retrieval-grounded operational objective: construct an executable neighborhood in Θ using a finite preset database, and then adapt within that neighborhood while enforcing feasibility. This yields a system whose intermediate artifacts (retrieved prototypes, draft parameters, constraint repairs) are auditable by design, which is critical for a journal setting emphasizing system rigor and reproducibility.

C. The Retrieval-Reasoning Decomposition

Because the inverse mapping from perceptual intent to parameters is ill-posed and the DSP validity constraints are strict, we implement a retrieval-grounded synthesis pipeline. Let z denote a retrieved preset (a discrete prototype) from a finite database. We first retrieve top-K candidates conditioned on q, then use the LLM to generate a draft parameter vector $\hat{\theta}$ conditioned on q and the retrieved context, and finally project the draft to the feasible set:

$$\tilde{\theta} = \Pi_\Theta \left(\hat{\theta} \right), \quad \hat{\theta} = \text{LLM}\left(q, \text{Top-}K(q) \right). \tag{3}$$

The projection operator Π_Θ enforces both range constraints and schema constraints (Section IV-D). Intuitively, retrieval supplies a coarse *mode-seeking* step that places the query in a plausible neighborhood of Θ, while the LLM performs a *local* adaptation step that refines parameters within that neighborhood under query-specific constraints.

a) Design Principle 1 (Well-defined synthesis under missing modalities).: Because $\mathbf{a}_{\text{ref}}$ is optional, the synthesis pipeline must remain well-defined when $\mathbf{a}_{\text{ref}} = \emptyset$. Our formulation treats q as a tuple with an explicit null element and requires every downstream stage (retrieval, fusion, and projection) to specify a deterministic fallback behavior when an input modality is absent (Section IV-C).

TABLE II: Positioning Against Prior Work

Work	Input	Output	Grounding	Editability
SAFE [6]	Text	Parameters	None	Yes
DDSP [3]	Audio	Parameters	None	Yes
LLM2Fx [17]	Text	Parameters	None	Yes
AudioLDM [1]	Text	Waveform	Pretrained	No
HM-RAG [28]	Text+Doc	Text	Retrieval	N/A
Audio-Agent (Ours)	**Text+Audio**	**Parameters**	**Dual-Modal RAG**	**Yes**

IV. Methodology

We implement the retrieval-grounded synthesis framework defined in Section III via a three-stage pipeline: (1) **Texture-Aware Retrieval** to retrieve candidate presets; (2) **Uncertainty-Guided Fusion** to combine multi-modal retrieval evidence; and (3) **Constrained Reasoning** to generate a draft $\hat{\theta}$ and project it into the valid set Θ.

A. Neuro-Symbolic Architecture

The system realizes the operator $\Phi(\mathbf{x}_{\text{in}}; \theta)$ through a decoupled design, visualized in Figure 1.

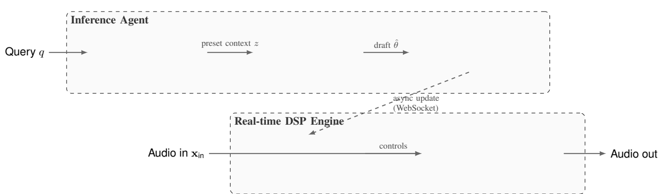

Fig. 1: The Neuro-Symbolic Architecture. The inference agent (top) retrieves candidate presets and adapts them with an LLM, while the deterministic DSP engine (bottom) executes real-time audio processing. The projection operator Π_Θ enforces plugin-defined validity constraints before parameters reach the real-time engine.

The architecture consists of three components. (i) **The real-time DSP engine** implements Φ as a low-latency audio processing graph and executes only validated parameters. We denote the parameter state consumed by the DSP engine as θ_{safe}, which can be smoothed over time to avoid zipper noise and audible artifacts. (ii) **The inference agent** implements the mapping $q \to \tilde{\theta}$ by running retrieval and LLM adaptation asynchronously at a lower rate than the audio callback. (iii) **The validity guard** implements the constraint function $\mathcal{C}(\theta)$ (Eq. 1) as a deterministic gatekeeper: any draft $\hat{\theta}$ must be mapped to $\tilde{\theta} \in \Theta$ before it is allowed to update θ_{safe}.

a) Design Principle 2 (Runtime isolation).: At inference time, the real-time audio path is a deterministic function of $(\mathbf{x}_{\text{in}}, \theta_{\text{safe}})$ through $\Phi(\mathbf{x}_{\text{in}}; \theta_{\text{safe}})$. The inference agent does not execute inside the real-time callback and can only influence audio through asynchronous updates of θ_{safe} after validity checking.

B. Texture Resonance Retrieval (TRR)

The core challenge in texture-aware retrieval lies in computing a similarity metric for the audio modality that preserves effect-relevant dynamics. Standard embeddings (e.g., mean-pooled Wav2Vec2 features) collapse temporal structure into a static vector, losing information that is critical for effects like modulation and distortion.

To address this, we introduce **Texture Resonance Retrieval (TRR)**. Drawing on the hypothesis that audio texture is encoded in second-order feature statistics [4], we compute the Gram matrix of deep activations.

Algorithm 1 Texture Resonance Retrieval (TRR) Encoding

Require: Audio sample a, Wav2Vec2 model $\mathcal{W}$
Ensure: Texture embedding $\mathbf{G}_{\text{norm}}$
1. Preprocessing: Normalize a and pad to minimum length.

2. Feature Extraction:
$\mathbf{H} \leftarrow \mathcal{W}_{\text{layer }\ell}(a) \in \mathbb{R}^{C \times T_{\text{frm}}}$
3. Gram Matrix Computation:
$\mathbf{G} \leftarrow \frac{1}{T_{\text{frm}}}\mathbf{H}\mathbf{H}^\top \in \mathbb{R}^{C \times C}$
4. Normalization:
$\mathbf{G}_{\text{norm}} \leftarrow \mathbf{G}/\|\mathbf{G}\|_F$ // *Frobenius norm*
return $\mathbf{G}_{\text{norm}}$

Let $\mathbf{H} \in \mathbb{R}^{C \times T_{\text{frm}}}$ be the activation map from an intermediate layer ℓ of a pre-trained Wav2Vec2 model, where C is the channel depth and T_{frm} is the temporal dimension. The texture descriptor $\mathbf{G} \in \mathbb{R}^{C \times C}$ is defined as the time-averaged correlation matrix (Algorithm 1):

$$\mathbf{G} = \frac{1}{T_{\text{frm}}}\mathbf{H}\mathbf{H}^\top \tag{4}$$

$\mathbf{G}_{ij}$ captures the co-activation of feature channels i and j irrespective of their absolute temporal position. This time-invariance allows TRR to match textures (e.g., "fast tremolo") even if the reference and database entries are not phase-aligned. We normalize $\mathbf{G}$ via the Frobenius norm for scale invariance: $\mathbf{G}_{\text{norm}} = \mathbf{G}/\|\mathbf{G}\|_F$. Similarity is then computed as cosine similarity between flattened $\mathbf{G}_{\text{norm}}$ vectors. Computationally, Gram-matrix construction scales as $O(C^2 \cdot T_{\text{frm}})$, which is more expensive than mean pooling ($O(C \cdot T_{\text{frm}})$); in our setting with short audio snippets, this overhead is manageable and can be further reduced via caching and approximate indexing.

a) Properties and limitations.: TRR can be viewed as a second-order summary of deep features: it retains cross-channel correlation structure while discarding absolute temporal alignment. This is desirable for retrieval in effect spaces where perceptual "style" is often expressed through repeated modulation patterns or stationary texture rather than a single time-locked event. The Frobenius normalization further reduces sensitivity to global scale changes in feature activations, making cosine similarity comparisons less dominated by magnitude. However, TRR is not a universal representation: by aggregating over time, it intentionally discards fine-grained ordering and can under-represent transient-dominant effects. Moreover, as with other deep representations, retrieval behavior can depend on the chosen backbone layer and the upstream feature extractor.

b) Lemma 3 (Alignment-insensitive texture statistics).: Let $\mathbf{H} \in \mathbb{R}^{C \times T_{\text{frm}}}$ be the frame-level feature matrix and let $\mathbf{P} \in \mathbb{R}^{T_{\text{frm}} \times T_{\text{frm}}}$ be any permutation matrix over frames. Then the TRR Gram matrix satisfies $\frac{1}{T_{\text{frm}}}\mathbf{H}\mathbf{H}^\top = \frac{1}{T_{\text{frm}}}(\mathbf{HP})(\mathbf{HP})^\top$. Therefore, TRR depends on time-aggregated co-activations and is insensitive to the absolute alignment of frames, which motivates its use as a texture-oriented retrieval signal.

C. Uncertainty-Aware Modal Fusion

The query q often contains asymmetric information. For example, a vague text prompt can produce a flat text-retrieval score distribution (high uncertainty), while the audio reference $\mathbf{a}_{\text{ref}}$ can still yield a peaked audio-retrieval distribution (low uncertainty). We propose a dynamic fusion strategy that weighs modalities by their *retrieval confidence*.

Algorithm 2 Uncertainty-Aware Modal Fusion

Require: Query $q = (t, \mathbf{a}_{\text{ref}})$, KBs $\mathcal{K}_{\text{text}}, \mathcal{K}_{\text{audio}}$, top-$K$ value K, hyperparameter β, small constant ϵ
Ensure: Fused candidates $\mathcal{R}$
  Text retrieval query: $q_{\text{text}} \leftarrow t$
  Audio retrieval query: $q_{\text{audio}} \leftarrow \mathbf{a}_{\text{ref}}$
  Retrieve top-K (text): $(\mathcal{R}_{\text{text}}, \mathbf{r}_{\text{text}}) \leftarrow \text{Search}(\mathcal{K}_{\text{text}}, q_{\text{text}}, K)$
  Normalize text scores to $[0, 1]$ (top-K min-max): $\mathbf{s}_{\text{text}} \leftarrow \text{Normalize}_{[0,1]}(\mathbf{r}_{\text{text}}; \epsilon)$
  if $\mathbf{a}_{\text{ref}} = \emptyset$ **then**
    $\mathcal{R}_{\text{audio}} \leftarrow \emptyset$, $\mathbf{s}_{\text{audio}} \leftarrow \mathbf{0}$, $w_{\text{text}} \leftarrow 1$, $w_{\text{audio}} \leftarrow 0$
  else
    Retrieve top-K (audio): $(\mathcal{R}_{\text{audio}}, \mathbf{r}_{\text{audio}}) \leftarrow \text{Search}(\mathcal{K}_{\text{audio}}, q_{\text{audio}}, K)$
    Normalize audio scores to $[0, 1]$ (top-K min-max): $\mathbf{s}_{\text{audio}} \leftarrow \text{Normalize}_{[0,1]}(\mathbf{r}_{\text{audio}}; \epsilon)$
    Convert to distributions: $\mathbf{p}_{\text{text}} \leftarrow \text{Softmax}(\mathbf{s}_{\text{text}})$, $\mathbf{p}_{\text{audio}} \leftarrow \text{Softmax}(\mathbf{s}_{\text{audio}})$
    Entropy: $u_m \leftarrow -\sum_i p_{m,i} \log p_{m,i}$ for $m \in \{\text{text}, \text{audio}\}$
    Normalized entropy: $\bar{u}_m \leftarrow u_m / \log K$
    Weights: $w_m \leftarrow \exp(-\beta \cdot \bar{u}_m)$, then normalize $w_{\text{text}} + w_{\text{audio}} = 1$
  end if
  Fuse scores: $s_{\text{fused}}(c) = w_{\text{text}} s_{\text{text}}(c) + w_{\text{audio}} s_{\text{audio}}(c)$ // *missing candidates treated as 0*
  $\mathcal{R} \leftarrow \text{Top-}K$ by s_{fused}
  return $\mathcal{R}$

a) Score calibration and preference distributions.: For each modality, we retrieve top-K candidates with raw similarity scores (e.g., cosine, dot-product, or BM25). Because score scales can differ across retrievers, any score-level fusion requires a calibration step; otherwise, the fused score can be dominated by an arbitrary scale rather than evidence. We apply a minimal calibration by normalizing the top-K scores into $[0, 1]$ via min-max normalization before computing the entropy of the resulting softmax distribution. Concretely, for a score vector $\mathbf{r}$ we compute $s_i = (r_i - r_{\min})/(r_{\max} - r_{\min} + \epsilon)$, where ϵ avoids division-by-zero when the top-K scores are nearly constant. We then convert the calibrated scores into a *normalized preference distribution* $p^{(m)}$ via softmax. Importantly, $p^{(m)}$ is not interpreted as a probabilistic posterior; it is a device for measuring how concentrated the modality's retrieval preference is over the top-K list. The modality weight is then computed via (Algorithm 2):

$$w_m \propto \exp\left(-\beta \cdot \frac{H(p^{(m)})}{\log K}\right) \tag{5}$$

where $H(p^{(m)})$ denotes the Shannon entropy of the modality-specific distribution $p^{(m)}$, K is the number of retrieved items, and β is a temperature hyperparameter. This assigns higher weight to modalities with "peaked" (low-entropy) distributions, effectively allowing the system to ignore ambiguous inputs. Larger β increases sensitivity to uncertainty: small entropy differences lead to more decisive shifts in w_{text} versus w_{audio}, whereas smaller β yields smoother, more conservative weighting.

b) Design Principle 4 (Entropy-weighted boundary behavior).: Under the calibrated score convention, the fusion mechanism exhibits the following deterministic behaviors: (i) if a modality produces an approximately uniform preference distribution over top-K, then its normalized entropy $\bar{u}_m$ is close to 1 and its weight decreases; (ii) if a modality produces a sharply peaked distribution, then $\bar{u}_m$ is small and its weight increases; (iii) if $\mathbf{a}_{\text{ref}} = \emptyset$, the algorithm reduces to a text-only pipeline with $w_{\text{text}} = 1$ by construction. These boundary cases make the fusion rule auditable and avoid undefined behavior when one modality is missing or uninformative.

We then rank candidates by the fused score and pass the top results as context to the LLM. For notational convenience in the experiments, we denote $\alpha \triangleq w_{\text{text}}$ and $1 - \alpha \triangleq w_{\text{audio}}$.

D. Constrained Reasoning and Projection

The LLM generates a draft parameter vector $\hat{\theta}$ based on the fused context. However, $\hat{\theta}$ is produced without hard constraints and is not guaranteed to lie within the valid set Θ.

We define a projection operator $\Pi_\Theta : \mathbb{R}^d \to \Theta$ that maps the draft to a valid configuration. In a plugin parameter space, validity is not only about numeric ranges, but also about *structural* constraints that encode domain logic. In our setting, these include: (i) **range constraints** (bounded sliders/knobs); (ii) **discrete constraints** (binary toggles and mode selectors); (iii) **conditional activity** (parameters that are only meaningful when a module is enabled); and (iv) **inter-parameter dependencies** (e.g., mutually exclusive modes or parameters that must be consistent with each other). We implement Π_Θ via two deterministic steps: **range clipping** and **schema enforcement**. First, for each dimension j we set $\theta_j \leftarrow \text{clip}(\hat{\theta}_j, \mathbf{l}_j, \mathbf{u}_j)$ to satisfy the box constraints. Then, we enforce $\mathcal{C}(\theta) = 1$ through deterministic repair rules, including discrete snapping (e.g., $\{0, 1\}$ toggles), dependency resolution, and bypass-aware gating (e.g., disabling parameters for bypassed modules to prevent "ghost" modifications). This design makes the system auditable: the LLM proposes a draft, while feasibility is guaranteed by a deterministic post-processing layer.

The final output $\tilde{\theta} = \Pi_\Theta(\hat{\theta})$ is executable by the DSP engine Φ by construction (it satisfies the plugin-defined constraints in Eq. 1).

a) Property 5 (Feasibility as an invariant).: By definition, Π_Θ maps any draft $\hat{\theta} \in \mathbb{R}^d$ to an element of Θ. Concretely, the range-clipping step ensures $\mathbf{l} \preceq \tilde{\theta} \preceq \mathbf{u}$, and the schema-enforcement step ensures $\mathcal{C}(\tilde{\theta}) = 1$. Therefore, every parameter state forwarded to the real-time engine is feasible and executable under the plugin-defined constraints.

E. Complexity and Latency Considerations

Journal readers often care not only about conceptual correctness but also about operational constraints. Our design separates (a) *real-time execution*, which must remain deterministic and low-latency, from (b) *asynchronous inference*, which can be slower and is not executed on the audio callback. On the inference side, TRR encoding dominates the audio branch with $O(C^2 T_{\text{frm}})$ Gram construction (Section IV-B). Candidate scoring is $O(Nd)$ for exact similarity over N candidates with feature dimension d; with ANN indexing, retrieval can be sublinear in N depending on index design and recall target. Fusion adds negligible overhead beyond top-K score normalization, softmax, and entropy computation (all $O(K)$). Projection is deterministic and scales approximately linearly with parameter dimensionality d plus finite schema-rule checks. End-to-end wall-clock latency is typically dominated by the LLM call, but this does not affect real-time determinism because inference runs asynchronously and updates θ_{safe} only after validity checking.

a) Reproducible latency protocol and current reporting boundary.: To quantify end-to-end control latency and "control lag" reproducibly, we define a five-stage timing breakdown: (i) TRR embedding (feature extraction + Gram), (ii) retrieval (top-K search), (iii) LLM adaptation, (iv) projection/repair, and (v) cross-layer update (e.g., WebSocket transport + parameter smoothing). A complete latency report should fix hardware, audio duration, and batch size, and report median and p95 over repeated warm-start runs. In this revision, we report the protocol and system decomposition only and intentionally defer numerical latency claims.

V. Experiments

We validate the proposed framework by answering three research questions: (RQ1) Does modeling second-order texture statistics (TRR) improve parameter inference compared to first-order baselines? (RQ2) Can the neuro-symbolic agent correct retrieval errors and satisfy constraints? (RQ3) Does uncertainty-aware fusion provide robustness against modality degradation?

A. Experimental Setup

We use a server-side benchmark of 1082 paired guitar-tone records (parameter JSON + rendered audio). Objective parameter-space evaluation uses a fixed 30-query held-out subset with ground-truth parameters; all remaining 1052 records form the retrieval knowledge base. The same 30-query protocol is used for fusion ablations and the reported Wav2Vec2 layer diagnostic.

a) Additional comparisons reported in supplementary material.: To strengthen auditability of our claims beyond the core RQ1–RQ3 ablations, we also report (i) a direct retrieval method comparison that includes **Pure LLM** generation without retrieval (to quantify the necessity of retrieval grounding), (ii) an LLM-enhanced retrieval ablation that tests whether few-shot ON/OFF prediction can correct module-level retrieval errors, and (iii) robustness stress tests under extreme modality degradation. In these supplementary experiments, retrieval is top-1, and the LLM baselines use `deepseek-chat` with 5 few-shot examples and no retrieval context for the Pure LLM condition. These are summarized in Supplementary Tables S1–S3.

b) Scope and statistical caution.: Although this revision reports a larger objective split ($N = 30$ with ground-truth parameters), we still interpret the results as evidence for a retrieval-grounded executable inference setting and avoid broad universal generalization claims.

Baselines. For reproducible texture-retrieval comparison, we evaluate **TRR** (Section IV-B) against two classical texture representations: **MFCC Temporal** and **Modulation Spectrogram**. For multimodal fusion (RQ3), we compare text-only and audio-only branches against entropy-weighted fusion using the same 30-query held-out subset. For method comparisons that include additional retrieval backbones (Text-RAG, Wav2Vec-RAG, FeatureNN-RAG) and generation-only baselines (Pure LLM), we report results in Supplementary Table S1.

B. Reproducibility and Evidence Traceability

To keep quantitative claims auditable, we report results under explicit evaluation protocols and metric definitions. The main paper reports RQ1–RQ3 evidence under a fixed server-side protocol (1082 corpus items with a 30-query held-out set; the remaining 1052 items form the retrieval knowledge base). Additional method comparisons and stress tests are reported as split-specific evidence in the supplementary material (Tables S1–S4) under their stated protocols and prompt settings. A detailed artifact manifest will be released with the accompanying code and data package upon publication.

C. RQ1: Efficacy of Texture Priors

Table III reports reproducible texture-retrieval alignment on the held-out subset.

Compared with classical texture baselines, TRR reduces parameter distance by 45.6% relative to MFCC temporal

TABLE III: Texture representation comparison on the held-out subset ($N = 30$). Lower is better for Param. Dist.; no universal generalization claim is made.

Method	Param. Dist. ↓	Acc@0.1 ↑	Cosine ↑
TRR (Ours)	13.57	0.734	0.973
MFCC Temporal	24.95	0.632	0.999
Modulation Spectrogram	32.02	0.651	0.998

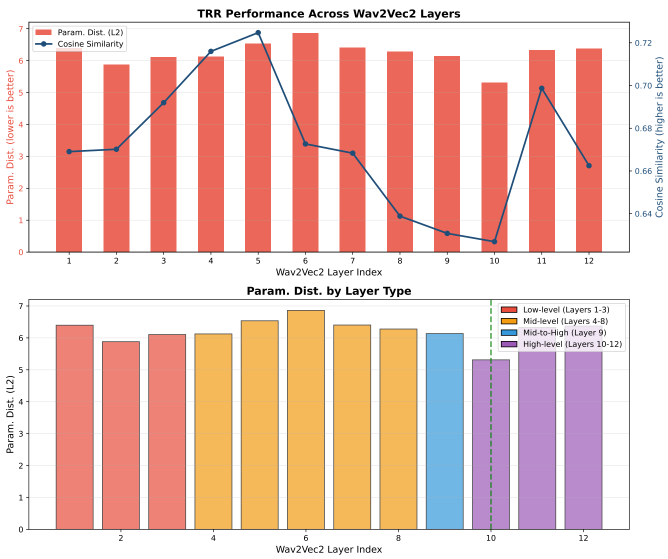

Fig. 2: Layer-wise parameter distance for TRR encoding ($n = 30$). In the current diagnostic output, layer 10 yields the lowest parameter distance among tested layers.

statistics and by 57.6% relative to modulation spectrogram under this benchmark, supporting the usefulness of second-order correlation structure for retrieval.

a) Necessity of retrieval augmentation.: To quantify the importance of retrieval grounding, we compared pure LLM generation (without RAG) against retrieval-grounded methods under a direct evaluation protocol (Supplementary Table S1, $N = 30$ queries). The pure LLM baseline fails catastrophically with Param. Dist.=13.3575, even when provided with few-shot examples. In contrast, TRR-based retrieval achieves Param. Dist.=0.3028, a 97.7% reduction in parameter error under the same operational metric. This dramatic gap demonstrates that LLMs alone cannot reliably generate accurate DSP parameters in this setting and that retrieval augmentation is essential for parameter-space accuracy.

b) Layer selection analysis.: TRR depends on intermediate representations from Wav2Vec2. We report a layer sweep diagnostic (Figure 2) with $n = 30$ queries. In the current diagnostic output, the best layer is 10, with several neighboring layers showing comparable order of magnitude. We use this sweep as an empirical sensitivity check rather than as a claim of universal layer optimality.

c) Complexity.: For a candidate set of size N and feature dimension d, exact retrieval requires $O(Nd)$ similarity computation per query, while ANN indexing can reduce empirical query time with index-dependent approximation trade-offs. Embedding construction is performed offline per audio and

TABLE IV: Computational complexity (informal). We report (i) the dominant per-audio embedding cost and (ii) the dominant per-query retrieval cost for a candidate set of size N and feature dimension d.

Method	Embedding cost	Retrieval cost
TRR (Gram)	$O(C^2 \cdot T_{\mathrm{frm}})$	$O(N \cdot d)$ (exact)
MFCC Temporal	$O(d \cdot T_{\mathrm{frm}})$	$O(N \cdot d)$ (exact)
Modulation Spectrogram	$O(d \cdot T_{\mathrm{frm}} \cdot \log T_{\mathrm{frm}})$	$O(N \cdot d)$ (exact)

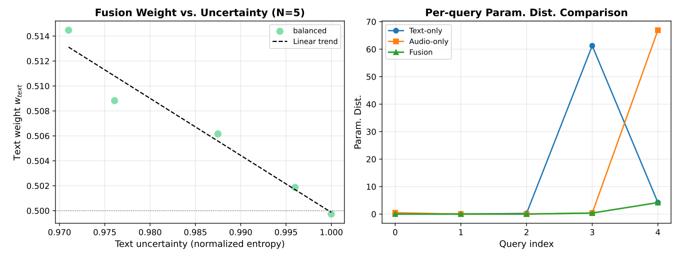

Fig. 3: Empirical fusion behavior on the held-out subset ($N = 30$): uncertainty-to-weight trend and per-query parameter-distance comparison.

can be cached.

D. RQ2: Neuro-Symbolic Correction

We evaluate the full pipeline ($\hat{\theta} \xrightarrow{\Pi_\Theta} \tilde{\theta}$) from a neuro-symbolic perspective, focusing on whether an LLM can *correct* retrieval imperfections rather than invent parameters from scratch. At the system level, retrieval provides executable prototype neighborhoods, LLM adaptation performs local query-conditioned edits, and deterministic projection Π_Θ enforces feasibility invariants (Section IV-D) before DSP execution.

a) Few-shot module correction evidence.: We report an LLM-enhanced retrieval ablation that isolates module-level correction (Supplementary Table S2, $N = 5$ core queries). Using 5 manually specified few-shot examples, the LLM is instructed to **ignore** retrieval-provided ON/OFF hints and predict only the effector activation pattern; we then copy numeric parameters from the retrieved reference conditioned on this predicted pattern. Under this protocol, TRR+LLM reduces Param. Dist. from 0.4104 (TRR retrieval-only) to 0.2631 and achieves Module=1.0, supporting the claim that the LLM can correct module-level retrieval errors when constrained by few-shot rules and exemplar parameters.

E. RQ3: Robustness via Uncertainty Fusion

To verify the adaptive weighting strategy (Section IV-C), we analyze reproducible fusion ablations and entropy-weight statistics on the held-out subset.

a) Fusion weight distribution and ablations.: We analyze the empirical fusion weights induced by entropy-based uncertainty and summarize both weight statistics and ablation performance (Text-only vs. Audio-only vs. Fusion).

Fusion yields a large relative gain over single-modality retrieval in this benchmark: compared to the best single modality (Text-only, Param. Dist.=13.15), fusion reduces parameter

TABLE V: Modal ablations on the held-out subset ($N = 30$). Results are reported with explicit scope boundaries and without universal generalization claims.

Method	Param. Dist. ↓	Acc@0.1 ↑	Cosine ↑
Text-only (TF-IDF)	13.15	0.837	0.705
Audio-only (TRR)	13.57	0.734	0.437
Fusion (Ours)	0.93	0.898	0.838

TABLE VI: Fusion weight statistics (beta=2.0) on the held-out subset ($N = 30$).

Weight	Mean	Std	Min	Max
w_{text}	0.506	0.006	0.500	0.514
w_{audio}	0.494	0.006	0.486	0.500

distance to 0.93 (92.9% reduction under the same metric definition). Because this effect size is estimated on $N = 30$, we report it as split-specific evidence and avoid broader universal generalization claims. We report a temperature sensitivity sweep in both the main paper and supplementary material.

b) Beta sensitivity.: We perform a sensitivity sweep over the temperature parameter β and observe that increasing β increases the variance of w_{text} while preserving overall performance in this benchmark.

c) Robustness under degraded modalities.: To validate the uncertainty-weighted fusion mechanism under degraded input conditions, we conducted stress tests simulating two extreme scenarios on a subset of core queries: (1) **text vague**, where text descriptions are replaced with generic terms (e.g., "warm"), and (2) **audio noisy**, where TRR vectors are corrupted with strong noise. In the text-vague condition, Text+LLM fails (Param. Dist.=29.28), while Hybrid+LLM with $\alpha = 0.15$ (15% text weight, 85% audio weight) maintains stable performance (Param. Dist.=0.26). In the audio-noisy condition, TRR+LLM fails (Param. Dist.=29.36), while Hybrid+LLM with $\alpha = 0.85$ maintains stable performance (Param. Dist.=0.28). Full metric breakdowns are reported in Supplementary Table S3.

F. Metrics

We employ five objective metrics to quantify alignment in Θ: (1) **Param. Dist.**: An operational parameter-space proxy computed as RMSE between flattened numeric parameter leaves, with missing keys treated as zero; (2) **Acc@0.1**: Fraction of flattened numeric parameters within ± 0.1 tolerance under the same operational representation; (3) **Recall**: Recall of active parameters; (4) **Cosine**: Angular similarity in parameter space; (5) **Module**: Jaccard index of active DSP modules. Detailed definitions are provided in the supplementary material. All reported parameter-distance values in this revision use the same **Param. Dist.** definition above. Legacy experiment logs sometimes refer to Param. Dist. as *L2* or *RMSE*; we use **Param. Dist.** consistently throughout. Because these metrics operate in parameter space, they are not perceptual audio distances. We therefore treat them as executable proxies for editability; additionally, for the listening-study similarity subset we report objective audio metrics as supplementary

TABLE VII: Fusion temperature sensitivity on the held-out subset ($N = 30$). Increasing β increases the variance of text weight w_{text} while maintaining stable parameter-distance performance in this benchmark.

β	$\mathbb{E}[w_{\text{text}}]$	Std	Param. Dist. (Fusion)
0.5	0.5016	0.0015	0.93
1.0	0.5031	0.0029	0.93
2.0	0.5062	0.0058	0.93
4.0	0.5124	0.0116	0.93

TABLE VIII: Trial 1 (style matching): system-level statistics (within-subject means).

System	Mean	Median	Std
Reference	86.85	92.50	16.70
Jazz Clean	76.19	81.00	20.80
Phase	75.38	80.50	20.88
Blues Solo	72.96	69.50	21.19
Ambient Guitar	72.35	73.50	18.45
Flanger	69.12	74.00	23.08
Modern Metal	65.58	70.00	30.22
Chorus	64.92	66.50	23.12

evidence (FAD_{vgg} [**? ?**], KL_{pann} [**?**], and CLAP_{scr} [36]; Supplementary Table S4).

Key Insight. Entropy-weighted modality balancing provides information complementarity in the reproducible ablation setting: text and audio branches each capture different retrieval cues, and uncertainty-aware weighting improves aggregate alignment over either branch alone.

G. Perceptual Listening Test (MUSHRA)

Protocol. We conduct a MUSHRA-style listening test with 26 participants, yielding 910 ratings across 10 trials after removing accidental header-like rows and placeholder entries (e.g., records where the email field contains the literal string "email"). Each trial uses a continuous 0–100 slider (higher is better). Following standard reporting practice, we group trials by task type and report (i) descriptive statistics over all scores and (ii) system-level statistics computed from within-subject means. Inferential comparisons use two-sided paired tests on within-subject means, and all aggregates are computed from the cleaned ratings and corresponding analysis outputs. In legacy experiment logs, the same system appears as **HCAP**; in this paper we use **Audio-Agent (Ours)** consistently.

1) Trial 1: Style Matching: **Data.** 208 ratings (26 participants $\times$ 8 systems). Overall mean 72.92 (median 77.50, std 22.72).

2) Trials 2–5: Guitar Solo Comparison: **Data.** 312 ratings (26 participants $\times$ 3 systems $\times$ 4 trials). Overall mean 69.53 (median 73.50, std 23.53). Audio-Agent (Ours) outperforms Manual tuning in this group: within-subject means are 71.55 (Audio-Agent) vs 51.72 (Manual), with a paired t-test yielding $p = 1.67 \times 10^{-6}$ ($n = 26$ participants).

3) Trials 6–10: Similarity to Reference: **Data.** 390 ratings (26 participants $\times$ 3 systems $\times$ 5 trials). Overall mean 57.54 (median 59.00, std 32.24). Audio-Agent (Ours) and MusicGen achieve comparable similarity scores: within-subject means

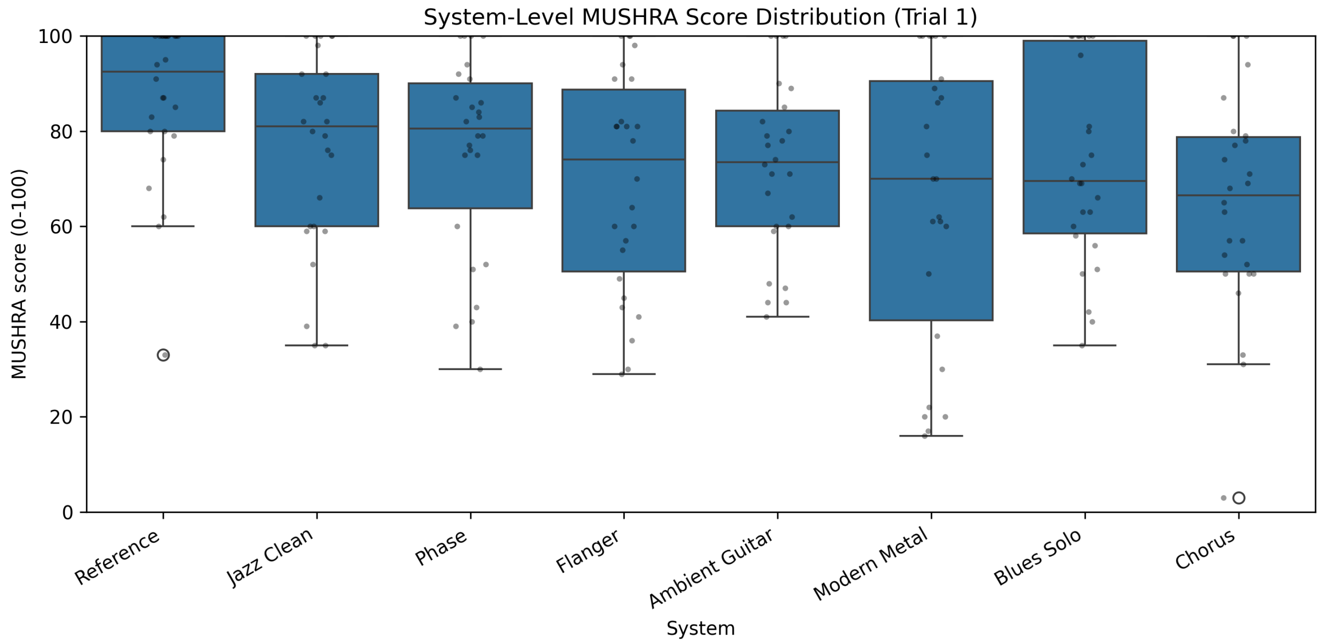

Fig. 4: Trial 1: system-level score distributions (boxplot) for style matching.

TABLE IX: Trials 2–5 (solo comparison): system-level statistics (within-subject means).

System	Mean	Median	Std
Reference	85.33	85.25	12.52
Audio-Agent (Ours)	71.55	72.88	12.41
Manual	51.72	55.00	17.15

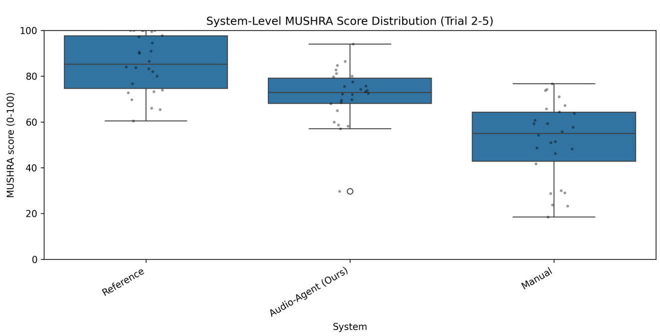

Fig. 5: Trials 2–5: system-level score distributions.

TABLE X: Trials 6–10 (similarity): system-level statistics (within-subject means).

System	Mean	Median	Std
Reference	89.02	91.50	10.04
Audio-Agent (Ours)	42.31	41.90	19.30
MusicGen	41.29	43.90	19.58

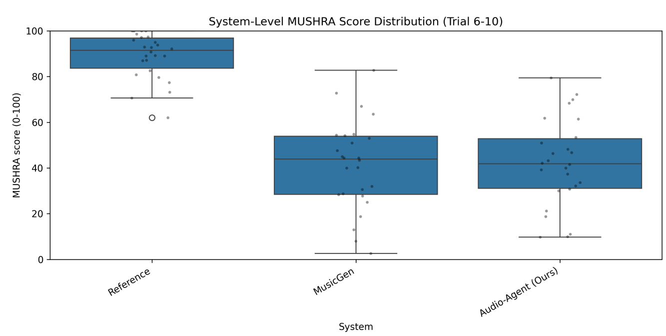

Fig. 6: Trials 6–10: system-level score distributions.

are 42.31 (Audio-Agent) vs 41.29 (MusicGen), while the Reference condition scores 89.02.

a) Objective audio metrics.: For the same five reference clips used in Trials 6–10, we additionally compute objective audio metrics against the reference: $\mathrm{FAD}_{\mathrm{vgg}}$ (VGGish-based Frechet Audio Distance) [**? ?**], $\mathrm{KL}_{\mathrm{pann}}$ (KL divergence over PANN label distributions) [**?**], and $\mathrm{CLAP}_{\mathrm{scr}}$ (text–audio similarity) [36]. As summarized in Supplementary Table S4, Audio-Agent achieves lower $\mathrm{FAD}_{\mathrm{vgg}}$ and $\mathrm{KL}_{\mathrm{pann}}$ than MusicGen (closer to the reference distribution under these proxies), while MusicGen achieves a slightly higher $\mathrm{CLAP}_{\mathrm{scr}}$ (stronger caption alignment).

b) Interpretation.: These results suggest that modality reliability should be treated as a first-class signal: when text collapses to generic intent, audio-conditioned retrieval can still constrain feasible parameter neighborhoods and stabilize generation.

VI. Discussion

A. Broader Implications

Audio-Agent suggests a practical pattern for grounded creative assistants: treat the LLM as a reasoning layer, while using retrieval over real presets to provide feasibility constraints and stylistic priors. This reduces hallucination and keeps outputs editable within standard production workflows. More broadly, it shifts the emphasis from open-ended generation to data-efficient adaptation, improving sample efficiency (reuse of real presets), interpretability (traceable references), and controllability (parameter-level editability).

B. Revisiting the Texture Hypothesis

Across the reported setup (Table III), TRR improves parameter-space alignment over classical texture baselines (MFCC temporal statistics and modulation spectrogram), supporting the hypothesis that second-order correlation structure can be useful as a texture descriptor for retrieval. At the same time, our objective benchmark is limited to a 30-query held-out subset with ground-truth parameters, and we do not claim that TRR defines a globally calibrated distance in parameter space. We therefore interpret TRR as a *retrieval prior* that can be beneficial for texture-sensitive control, rather than a universal metric for parameter transfer.

An open direction is whether analogous second-order representations help bridge modality gaps in other continuous control settings (e.g., video textures, haptic signals, or physiological time series), where the "style" of a signal may be expressed through co-activation structure rather than pointwise magnitudes. Future work can validate this hypothesis via controlled ablations on non-stationarity and by pairing retrieval with human preference judgments.

C. Ethical Considerations

Audio-Agent is intended to augment user creativity rather than replace it. Risks include homogenization toward common exemplar neighborhoods and over-reliance on suggested settings. In deployment, the interface should emphasize transparency (e.g., showing retrieved references) and user agency (e.g., direct editability and opt-out of retrieval). More generally, any model or retrieval corpus built from existing recordings or presets should consider copyright, licensing, and data provenance constraints.

VII. Conclusion

We presented Audio-Agent, a neuro-symbolic system for generating editable DSP parameters from natural language descriptions with texture-aware retrieval and LLM-based correction. Across reproducible objective metrics (Table III) and fusion ablations (Table V), retrieval grounding with uncertainty-aware fusion improves parameter alignment under the reported benchmark settings.

Looking forward, we envision expanding preset coverage for under-represented effect families and integrating controlled listening tests to validate perceptual quality. We also aim to generalize beyond guitar effects to broader production parameter spaces, including mixing and mastering workflows.

We hope this approach lowers the barrier to expressive sound design and broadens creative access to professional-grade production control.

Future Work

Our evaluation uses a 30-query held-out set from a 1082-record corpus. Future work should validate on larger datasets ($N \geq 100$) with k-fold cross-validation and bootstrap confidence intervals to strengthen statistical robustness. While we compare against classical texture representations (MFCC, modulation spectrogram), comparison with modern audio embeddings (CLAP, AST, HuBERT) remains to be done.

Audio-domain metrics (Fréchet Audio Distance, MR-STFT) and parameter-perceptual correlation analysis would strengthen the connection between parameter-space optimization and perceptual quality. The many-to-one nature of timbre—where multiple distinct parameter configurations yield perceptually similar sounds—requires controlled listening tests to establish perceptual ground truth and calibrate objective metrics against human judgment.

Cross-domain generalization to other instruments and effect types remains an open direction, as does comprehensive latency profiling with reproducible protocol definitions for real-time deployment claims. Hybrid approaches that combine retrieval-based generation with targeted parameter optimization could expand the reachable sound space while preserving editability.

Active learning strategies that identify database gaps and user feedback loops could improve coverage for under-represented effect families (e.g., reverse reverbs, granular synthesis). For fusion reliability, cross-modal consistency checks and fallback mechanisms for conflicting modalities would strengthen robustness when text and audio inputs disagree.

Acknowledgments: We thank the anonymous reviewers for their valuable feedback.

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